

Date: 20 Jan 2025

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Power Electronics

Lecture 1

Converter: Change DC voltage from one level to another

* Transformer change AC voltage from one level to another

* Step up voltage

* Generating nothing so reduce something

* We can change 10V DC to 100V DC using power electronics by using converter

* We can step up DC voltage

* We can step down DC voltage

→ Boost Converter: If 10V DC is step up to 100V DC then it is a boost converter

→ Buck converter: If 100V DC is step down to 10V DC then it is a buck converter

4 Scenarios

Case 1 Conversion of AC to DC

→ This process is called rectification

→ This system is called rectifier

Example: mobile phone charger (there is no DC in homes convert AC to DC)

* Kitchen appliances is DC (DC appliances has their own inverter)

Case 2 Conversion of DC to AC

→ This system is called inverter

Example: UPS (convert DC to AC)

→ When electricity UPS charges battery convert AC to DC (like mobile phone but with higher rating)



→ When no electricity it need to run load (all load is AC fan, lights stored energy in battery is converted into AC
↳ battery is always DC

KABIRI
NOTE

Case 3 Conversion of DC to DC

- We can step up and step down DC
- These circuits are called Choppers / DC-DC converters
- They are use in supplies, switch modes, boost converters
- One is on at one time

Case 4 Conversion of AC to AC

- Here Frequency is varied
- These are called cyclon converters
- cyclons are like due to cycle of frequency
- Their purpose are not to change level of DC then can be done by transformer not by power electronic here frequency is varied

Why frequency is varied?

Frequency change mean speed is change

- Low frequency mean low speed
- High frequency mean high speed.

Power Electronic course:

This course is for:

Study, design and applications of electronic circuits and device to effectively control and convert electrical power

Applications

→ Small signals are modified using op amplifiers

* To drive large motors

→ op amplifier (high gain electronic voltage amplifier)

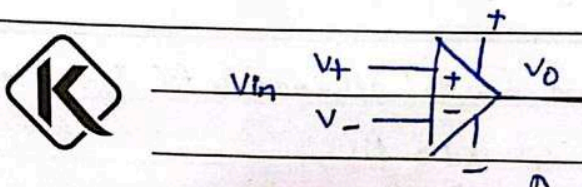
* Robotic

Input impedance (very high)

* Renewable energy

* Power supplies

* Electronic vehicles



- * Input current is zero (due to high Impedance)
- * Open loop (difference b/w V_+ and V_- is 0 by high Impedance)

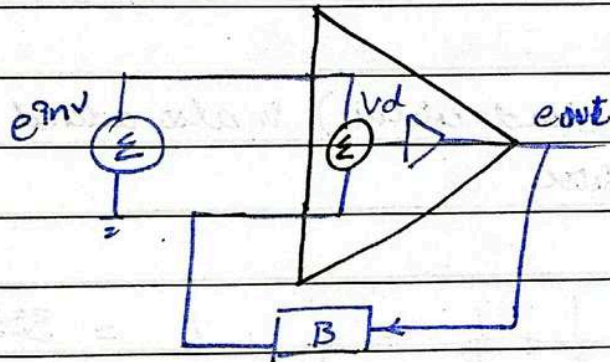
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→ We don't rely on open loop op amplifier as output is dependent on gain

If input difference is more than there large change in output (like just 1mV difference will result double output)

To limit this output we will connect feedback. We will connect feedback to negative (inverting terminal)



NI → non inverting
 IN → inverting

$$e_{inv} \neq V_d = e_{out}$$

$$V_d = e_{ni} - e_{inv} \quad \therefore e_{ni} = e_{in}$$

$$0 = e_{in} - B e_{out}$$

$$e_{in} = B e_{out}$$

Negative feedback

↳ Inverting Input

↳ Input current is zero (due to high impedance)

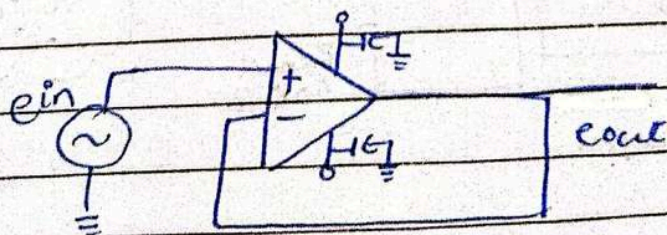
↳ $V_+ = V_-$

Unity Feedback / Voltage follower:

↳ $e_{out} = e_{in}$

↳ gain 1 (gain & output over input)

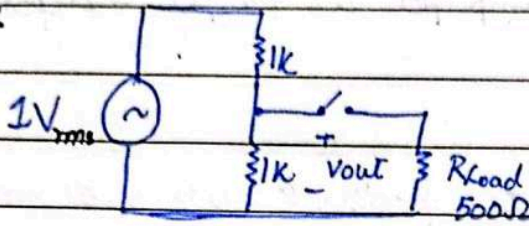
Benefit: If we place load after this and changes load due to which current changes current will not flow through inside of amp pass through feedback but will provide the required output like "buffer".



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Example 1

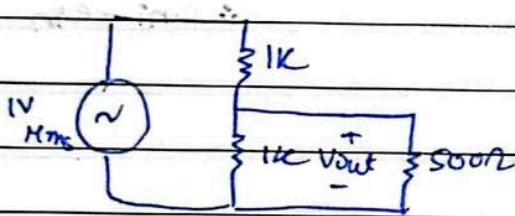


When switch is open so

$$V_{out} = \frac{1k \times 1}{1k + 1k} = 0.5V$$

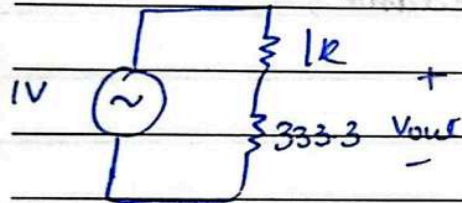
$$|V_{out} = 0.5V|$$

When switch is close so the 500Ω (load resistor) is also added up forming a parallel combination with 1k resistor



$$R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{1}{\frac{1}{1k} + \frac{1}{500}} = 333.3\Omega$$

$$|R_{eq} = 333.3\Omega|$$



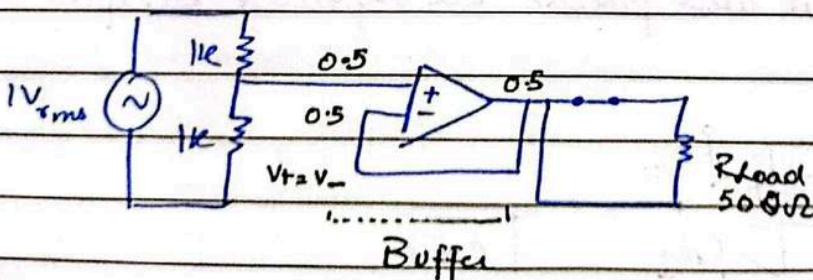
$$V_{out} = \frac{333.3}{1k + 333.3} \times 1 = 0.25V$$

$$|V_{out} = 0.25V|$$

Now it will draw double current
 power will not change

Load change voltage reduce power high

We want to place load on/off it but no voltage change take place so we will use op amp



Source Degradation:

Load change Karray
 Sa supply reduce
 hoti

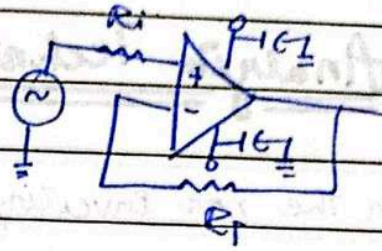


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Non Inverting Amplifier:

$$A_{cl} = 1 + \frac{R_f}{R_i}$$



Example

$$R_f = 10K \quad R_i = 1K \quad V_{in} = 0.5$$

$$A_{cl} = 1 + \frac{R_f}{R_i}$$

$$\frac{V_o}{V_i} = \left(1 + \frac{R_f}{R_i} \right) \Rightarrow V_o = V_i \left(1 + \frac{R_f}{R_i} \right)$$

$$V_o = 0.5 \left(1 + \frac{10K}{1K} \right) = 0.5(1+10)$$

$$\boxed{V_o = 5.5V}$$

Example 2 Design a non inverting amplifier with gain of 100

$$R_f = 100K$$

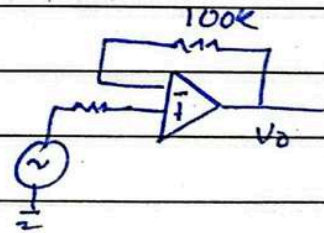
$$A_{cl} = 1 + \frac{R_f}{R_i}$$

$$\frac{R_f}{R_i} = (A_{cl} - 1)$$

$$R_i = \frac{R_f}{A_{cl} - 1} = \frac{100K}{100 - 1}$$

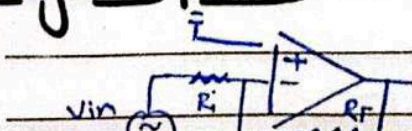
$$\boxed{R_i = 1.01K \Omega}$$

as $1.01K \Omega$ is not practically available so we will use $1K \Omega$ resistor that will impact its gain.



Inverting Amplifier:

$$A_{cl} = -\frac{R_f}{R_i}$$



$$V_{in} = 0.5 \quad R_f = 100K \quad R_i = 10K$$

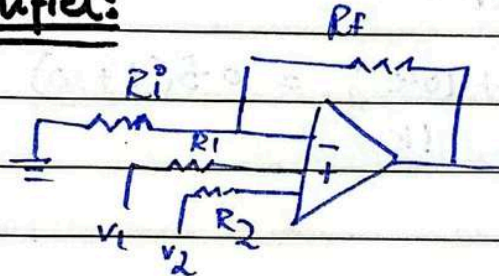
$$V_o = -\frac{R_f}{R_i} V_{in} \Rightarrow V_o = -0.5 \times \frac{100K}{10K} \Rightarrow \boxed{V_o = -5V}$$

KARIM'S
NOTE

Universal Analysis Technique:

- Voltage on the non inverting input pin
- Voltage on the inverting input pin
- Current Through each of the input resistors
- KCL → feedback current
- Voltage drop across feedback resistor
- Output voltage using KVL

Summer Amplifier:



$$V_{out} = \left(1 + \frac{R_F}{R_1}\right) \left(\frac{R_2}{R_1 + R_2} \times V_1 + \frac{R_1}{R_2 + R_1} \times V_2 \right)$$

Example 3

$$E_{out} = \frac{-10}{470} \times 15 - \frac{10}{5.6} \times 1 = -2.104$$

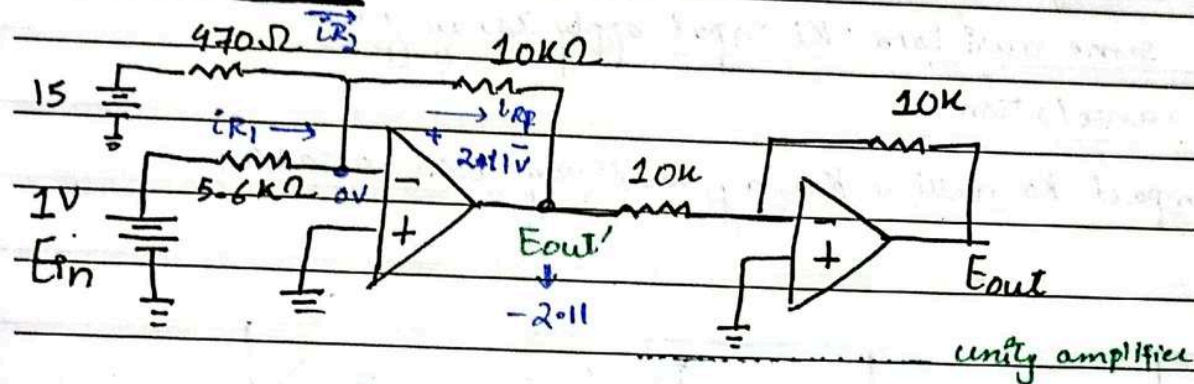
$$E_{out} = -1(-2.104)$$

$$\boxed{E_{out} = 2.104V}$$



Lecture 2

Example 3



Step 1 voltage at non inverting terminal (V_+)

$$V_{N+} = 0V$$

Step 2 voltage at inverting terminal (V_-)

$$V_{INV} = 0V$$

Step 3 current through input resistance

$$i_{R_1} = \frac{1-0}{5.6k} = 179\mu A$$

$$i_{R_2} = \frac{15-0}{470} = 32\mu A$$

Step 4 KCL $\rightarrow$ feedback current

$$i_{R_f} = i_{R_1} + i_{R_2} = 179\mu A + 32\mu A$$

$$i_{R_f} = 211\mu A \rightarrow$$

We will apply loop

$$\begin{array}{c} \text{for } E_{out}' \\ \begin{array}{|c|} \hline + \\ \hline 2.11V \\ \hline + \\ \hline E_{out}' \\ \hline - \\ \hline -E_{out}' - 2.11 + 0 \\ \hline \end{array} \\ \hline E_{out}' = -2.11V \end{array}$$

Step 5 voltage across feedback

$$V_{R_f} = i_{R_f} \times R_f = (211\mu A)(10k)$$

$$V_{R_f} = 2110mV = 2.11V$$

Step 6 for output voltage

$$\text{As non inverting } G = -\frac{R_2}{R_1} = -\frac{10}{10} = -1$$

$$E_{out} = -1(-2.11) = 2.11V$$

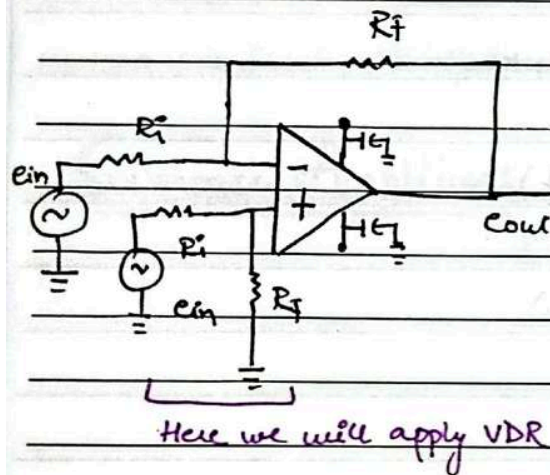


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Difference Amplifier:

- ↳ mismatch b/w two sources (2 sources ka b/w karna difference wo balana)
- ↳ Calibration (aik signal a kaha ha usko offset karna, zero karna same auri tara ki input apply karay ga)
- ↳ use for noise cancelation
- ↳ kisi cheez ka impact ko nullify karay ka liyay use ho sakta



$$e_{out} = \frac{R_f}{R_i} (e_{in+} - e_{in-})$$

↳ only possible when there is same value of resistor

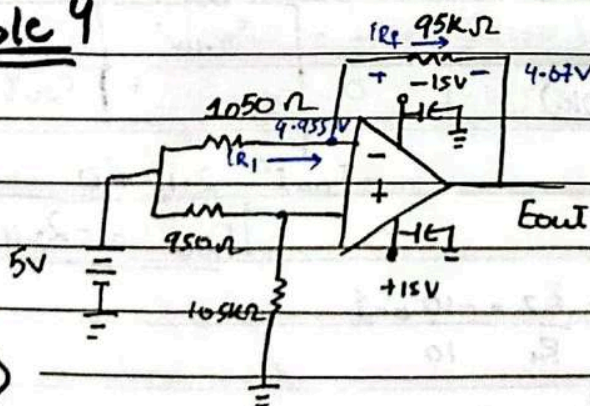
$$e_{out} = \underbrace{-\frac{R_f}{R_i} e_{in-}}_{\text{Inverting terminal}} + \underbrace{\left(1 + \frac{R_f}{R_i}\right) \left(\frac{R_f}{R_f + R_i}\right) e_{in+}}_{\text{Non inverting terminal}}$$

$$e_{out} = -\frac{R_f}{R_i} e_{in-} + \left(\frac{R_i + R_f}{R_i}\right) \left(\frac{R_f}{R_f + R_i}\right) e_{in+}$$

$$= -\frac{R_f}{R_i} e_{in-} + \frac{R_f}{R_i} e_{in+}$$

$$e_{out} = \frac{R_f}{R_i} (e_{in+} - e_{in-})$$

Example 4



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Step 1 voltage at non inverting terminal (V_+)

$$V_{N1} = \frac{105(K)}{105K + 950} \times 5$$

$$V_{N1} = 4.955V$$

Step 2 voltage at inverting terminal (V_-)

$$V_{INV} = 4.955V \Rightarrow V_+ = V_-$$

Step 3 Current through each of the input resistor

$$I_{R1} = \frac{5 - 4.955}{1050}$$

$$I_{R1} = 42.86 \mu A \rightarrow$$

Step 4 Current through feedback resistor

$$I_{Rf} = 42.86 \mu A$$

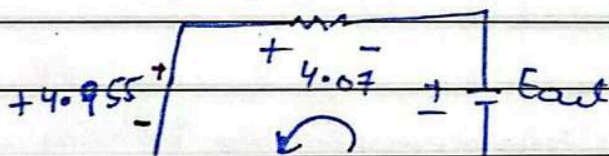
Step 5 Voltage across feedback resistor

$$V_{Rf} = I_{Rf} R_f = (42.86 \mu A)(95K\Omega)$$

$$V_{Rf} = 4.07V \quad + \text{ to } - \quad +4.07V$$

Here as negative feedback so we are calling input to inverting terminal and same current will pass through its feedback

Step 6 Output voltage



$$\begin{aligned} -4.955 + 4.07 + E_{out} &= 0 \\ E_{out} &= 0.885V \end{aligned}$$

$$-E_{out} - 4.07 + 4.955 = 0$$

$$\begin{aligned} E_{out} &= 0.885V \\ E_{out} &= 885mV \end{aligned}$$

b) $R_{f1} = 1050\Omega$ $R_{i2} = 950\Omega$ $R_{f1} = 95K$ $R_{f2} = 105K\Omega$

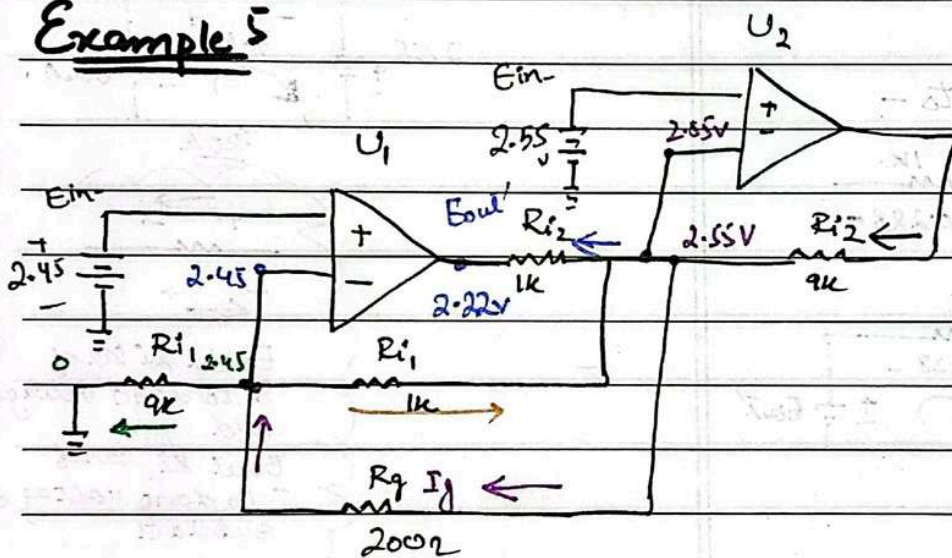


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↳ due to mismatch resistor of difference amplifier we replace it with instrumentation amplifier moreover input cannot be same

↳ Gain adjustable resistor

Example 5


$$V_{N1} = 2.45$$
$$V_{INV} = 2.45 \quad | \quad N_4 = V_-$$
$$i_{R_1} = \frac{2.45}{9k} = 272.2 \mu A$$
$$I_{R_1} = 272.2 \mu A$$
$$I_g = \frac{2.55 - 2.45}{200} = 500 \mu A$$
$$I_g = 500 \mu A$$
$$V_{N1} = 2.55V$$

$V_{INV} = 2.55V$	$V_+ = V_-$
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$$I_{R2} = \frac{2.55 - 2.22}{1k} = 328 \mu A$$
$$I_{R2} = 328 \mu A \quad \leftarrow$$

Step 4 current through feedback (9k)

$I_f = 328 \mu A + 500 \mu A$

$I_f = 828 \mu A$ ←

↳ KCL KARIMI
current NOTES
entering = sum of
currents leaving

Like KARIMIA
current NOTES
entering = Sum of
current Leaving

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huc 500 μ A is entering in the node and 272.2 μ A is leaving the node

Step 4 current through feedback (1k)

$$I_{R_f} = 500_{\mu} - 272.2_{\mu}$$

$$I_{R_f} = 227.8 \mu A \rightarrow$$

Step 5 voltage across feedback

$$V_{R_f1} = I_{R_f1} (R_{f1})$$

$$= (228 \mu A) (1k)$$

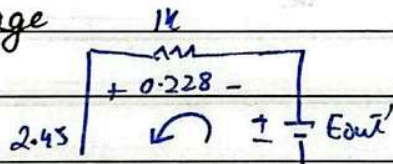
$$V_{R_f1} = 228 mV$$

$$V_{R_f1} = 0.228 V \quad + \text{ to } -$$

$$\begin{array}{c} 1k \\ \text{---} \\ + 0.228 - \end{array}$$

Step 6 output voltage

$$E_{out}' = ?$$



$$-E_{out}' - 0.228 + 2.45 = 0$$

$$E_{out}' = 2.22 V$$

Step 5 voltage across feedback (9k)

$$V_{R_f2} = I_{R_f2} R_{f2}$$

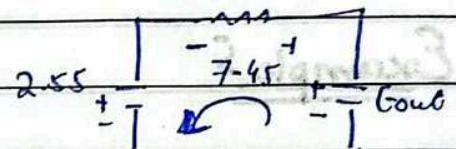
$$= (828 \mu A) (9k)$$

$$V_{R_f2} = 7.45 V$$

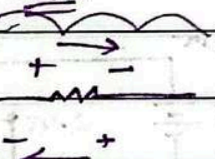
Step 6 output voltage

$$2.55 + 7.45 - E_{out} = 0$$

$$E_{out} = 10 V$$



Trace

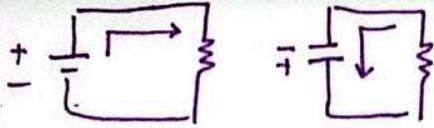


E_{out} ki taraf
+ to dono voltage
add

E_{out} ki taraf
- to dono voltage
subtract

b) $R_g = 400 \Omega$





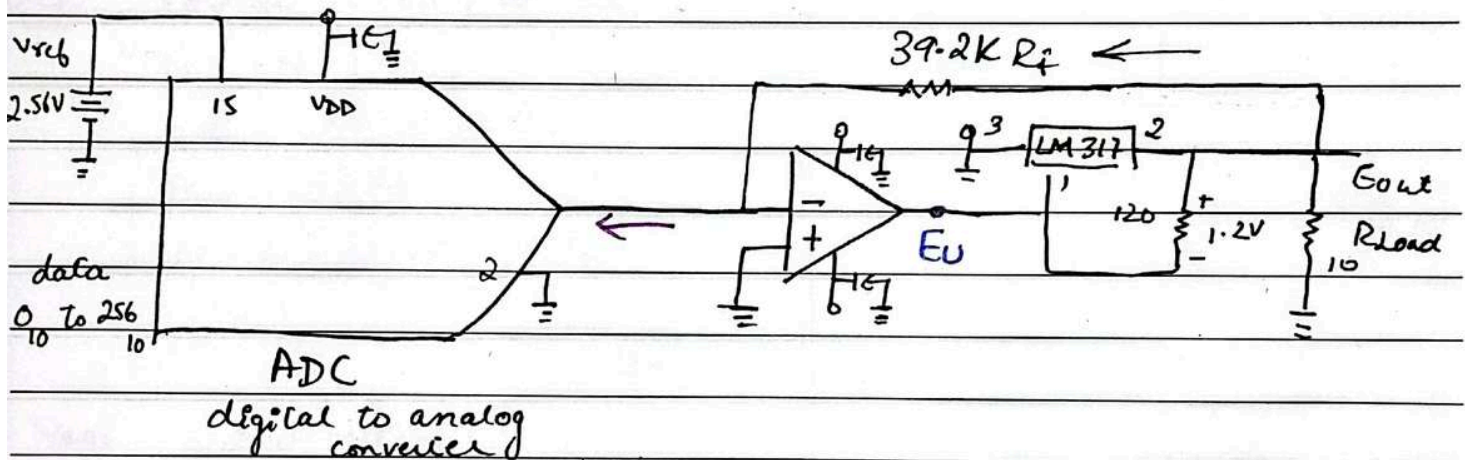
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voltage follower does not degrade voltage

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Composite Amplifier:

- ↳ to overcome weakness and to amplify strength
- ↳ composes of several amplifier
- ↳ necessary to have IC in feedback path to form composite amplifier if only resistor than normal amplifier



ADC

digital to analog converter

$$I_{out} = \frac{V_{ref}}{10k\Omega} \times \frac{code}{256_{10}}$$

Step 1 voltage at non inverting terminal (V_+)
 $V_{NH} = 0V$

Step 2 voltage at inverting terminal (V_-)
 $V_{INV} = 0V$ $V_+ = V_-$

Step 3 current through input resistance

$$I_{out} = \frac{V_{ref}}{10k\Omega} \times \frac{code}{256_{10}}$$

$$= \frac{2.56}{10} \times \frac{128}{256}$$

$$I_{out} = 128\mu A \leftarrow$$

Step 4 current through feedback resistor (39.2K)
 $I_F = 128\mu A$

Step 5 voltage across feedback resistor

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$$V_{Rf} = I_{Rf} R_f$$

$$= (128 \mu) (39.2 k\Omega)$$

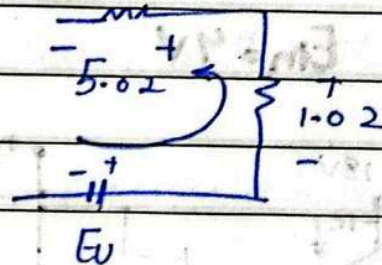
$$V_{Rf} = 5.02V$$

Step 6 output voltage
 $E_{out} = 5.02V$

output voltage at op amp

$$-E_u + 5.02 - 1.02 = 0$$

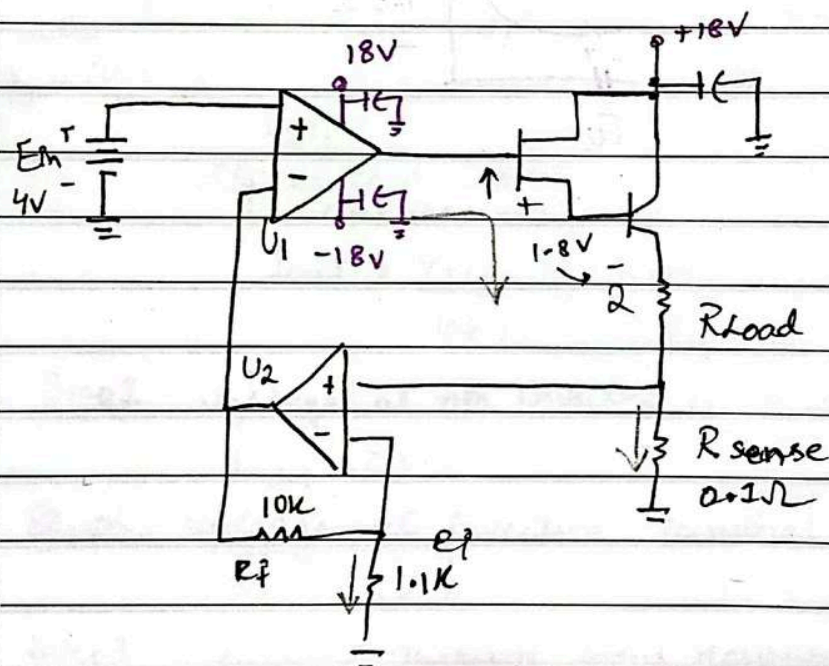
$$E_u = 3.82V$$



Lecture 3:Active feedback Composite Amplifier:

- ↳ only passive elements like resistor normal feedback
- ↳ With passive active element in feedback path like inductor

Work as drive, Low price provide high current (5A)

Example 7 $E_m = 4V$ Step 1

$$V_+ = 4V$$

Step 2

$$V_- = 4V$$

$$V_+ = V_-$$

Step 3

$$I_{R_f + R_i} = \frac{4}{10k + 1k} = 360.36 \mu A$$

$$I_{R_f + R_i} = 360.36 \mu A$$

Step 1

$$V_{INV} = (360.36 \mu)(1.1k) = 0.396$$

$$|V_{INV}| = 396.4 mV$$

Step 2

$$V_{N1} = 396.4 mV$$

Step 3

$$I_{R_{sense}} = \frac{396.4 mV}{0.1} = 3.964 A$$

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$$I_{R_{sense}} = 3.964A$$

$$V_{emitter} = 3.964(0.1 + 2)$$

$$V_{emitter} = 8.324V$$

$$E_{out} = 8.324 + 1.8$$

$$E_{out} = 10.124V$$

Suspended Supply Amplifier:

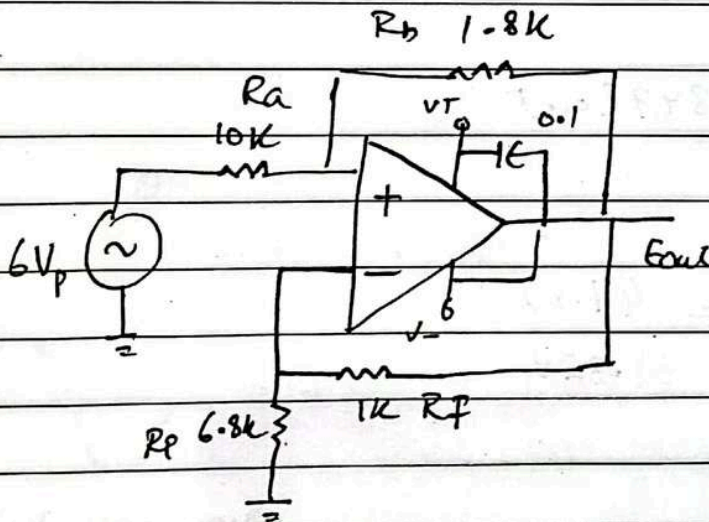
- ↳ supply voltage will varied (within the range)
- ↳ Minimal headroom voltage
- ↳ Allow to reduce unnecessary power dissipation

↳ Heating is due to power efficient

Example 8

$$V_{margin} = 8V$$

↳ A efficient system will not be heated



Step 1

$$V_{inv} = \frac{6.8K}{6.8K + 1K} \times E_{out}$$

$$V_{INV} = 0.871 E_{out}$$



Step 2

$$V_{NI} = 0.871 E_{out}$$

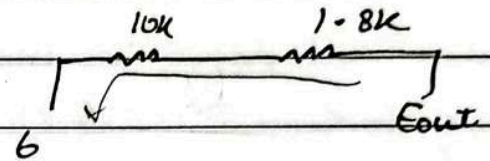
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Step 3

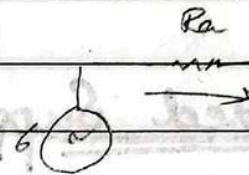
$$I_{Ra} = \frac{6 - 0.87 E_{out}}{10} = 0.6 - 0.087 E_{out}$$



$$I_a = \frac{E_{out} - 6}{1.8k + 10k}$$

$$V_{supply} = \frac{V_{in}}{2} = 3V$$

$$I_a = 84.74 \mu E_{out} - 508.47 \mu$$



$$6 + I_a R_a = V_{N1}$$

$$6 + (84.7 \mu E_{out} - 508.47 \mu) 10k = V_{N1}$$

$$\frac{6 - E_{out}}{(10 + 1.8)k}$$

$$6 + 847 \times 10^{-3} E_{out} - 5084 \times 10^{-3}$$

$$V_{N1} = 6 - I_a R_a$$

$$6 + 0.847 E_{out} - 5.084$$

$$V_{N1} = 0.916 + 0.847 E_{out}$$

$$V_{N1} = V_{N1}$$

$$0.87 E_{out} = 0.916 + 0.847 E_{out}$$

$$E_{out} = 39.82V$$

$$V^+ = 39.82 + 8 = 47.82V$$

$$V^- = 39.82 - 8 = 31.82V$$



Power Parameter Calculation

If signal is provided Then we have to calculate
average value

Rms

Power delivered

DC waveform

Simplest waveform

Constant value

$$V(t) = V_p$$

V_p

0

Example

- Batteries
- Mobile charger input

Sinusoidal waveform

$$V(t) = V_p \sin \omega t$$

or

$$V(\theta) = V_p \sin \theta$$

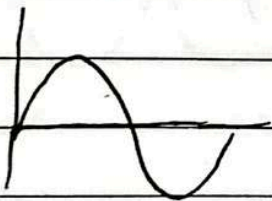
$$\theta = \omega t$$

Angular frequency

T → Time period

$$T = \frac{1}{f}$$

$$\omega = 2\pi f$$



Inverted sine wave

$$V(t) = -V_p \sin \omega t$$

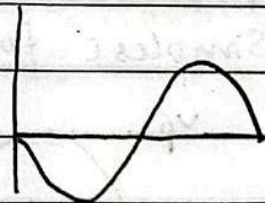
or

$$V(\theta) = -V_p \sin \theta$$

or

$$V(\theta) = -V_p (\sin \theta + \pi)$$

$$V(\theta) = -V_p (\sin \theta - \pi)$$



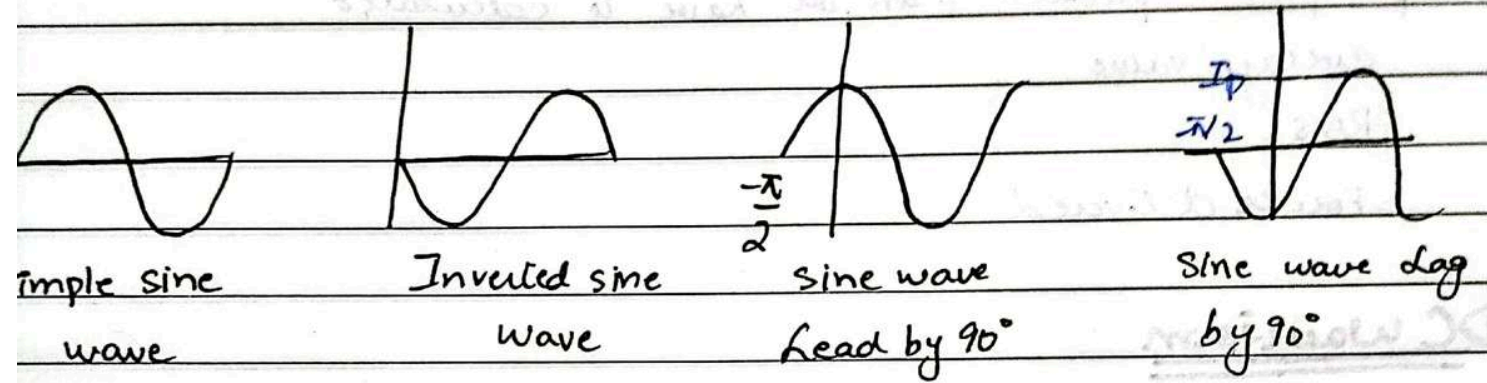
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Effect of capacitor and inductor

Inductor $\rightarrow$ current lag

Capacitor $\rightarrow$ current lead



Purely capacitive circuit

$$i(\theta) = I_p \sin\left(\theta + \frac{\pi}{2}\right)$$

here current lead the voltage by 90°

Purely inductive circuit

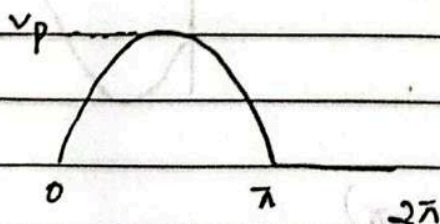
$$i(\theta) = I_p \sin\left(\theta - \frac{\pi}{2}\right)$$

here current lag the voltage by -90°

half wave Rectified Sine wave

AC To DC conversion

Simplest form : Single diode



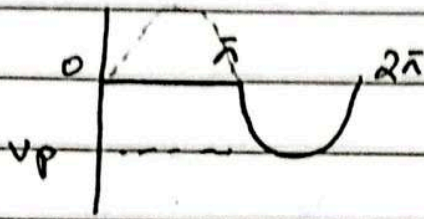
$$V(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta \leq \pi \\ 0 & \pi \leq \theta \leq 2\pi \end{cases}$$



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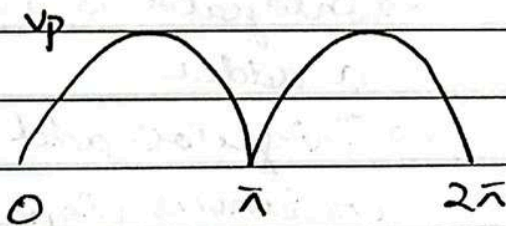
$$V(\theta) = \begin{cases} 0 & 0 \leq \theta \leq \pi \\ -V_p \sin \theta & \pi \leq \theta \leq 2\pi \end{cases}$$

Full Wave Rectified Sine wave

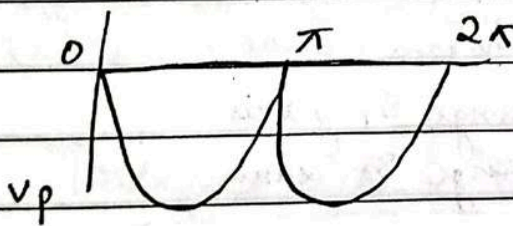
→ AC to DC conversion

→ Simplest form two diodes with center tapped transformer

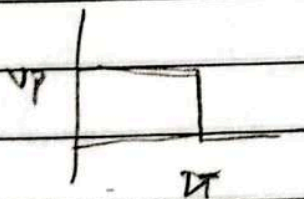
→ Second part of waveform is inverted



$$V(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta \leq \pi \\ -V_p \sin \theta & \pi \leq \theta \leq 2\pi \end{cases}$$



$$V(\theta) = \begin{cases} -V_p \sin \theta & 0 \leq \theta \leq \pi \\ V_p \sin \theta & \pi \leq \theta \leq 2\pi \end{cases}$$



$$V(\theta) = \begin{cases} V_p & 0 \leq t \leq DT \\ 0 & DT \leq t \leq T \end{cases}$$



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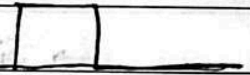
Lecture 4

Rectangular Wave:

- use for switching (ON for sometime off for sometime)

Duty Cycle $D = \frac{t_{on}}{T}$

$$v(t) = \begin{cases} V_p & 0 < t < DT \\ 0 & DT < t < T \end{cases}$$

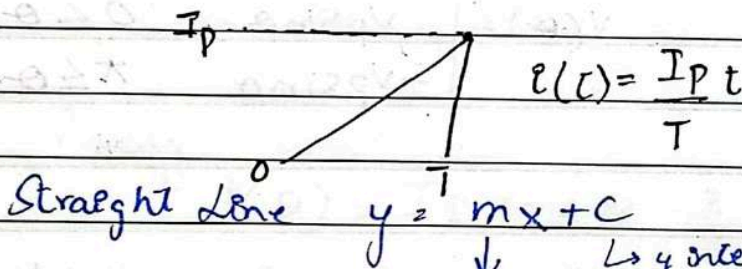


Triangular Wave:

→ Capacitor work as integrator

→ Integrator is actually a adder

→ Integrator add continuous signal



Slope

$m = \text{slope} = \frac{\Delta y}{\Delta x} = \frac{\text{change in y axis}}{\text{change in x axis}}$

Change in y axis $\leftarrow m$ $\leftarrow$ Change in x axis

$i(t) = \left(\frac{I_p}{T} \right) t + 0$ y intercept (initial value)

Change in x axis $\leftarrow$ function

$i(t) = \frac{I_p t}{T}$

Example 1

Ramp signal peak current 3A frequency 150 kHz

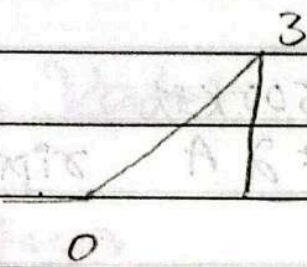
Calculate current 5 μ s after beginning of signal



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$$f = 150 \text{ kHz}$$

$$T = \frac{1}{f}$$

$$T = 6.67 \mu\text{s}$$

$$i(t) = \frac{I_p}{T} t + 0$$

$$i(t) = \frac{3}{6.67 \mu\text{s}} t$$

$$t = 5 \mu\text{s}$$

$$= \frac{3}{6.67 \mu\text{s}} (5 \mu\text{s})$$

$$i(t) = 2.248$$

sinewave is cyclic
 so we don't need
 to worry about
 time

28 Rec (here in ramp we
 need to take care
 that it will)

$$10 \mu\text{s}$$

$$t = 10 \mu\text{s} - 6.67$$

$$t =$$

$$t = 15 \mu\text{s} - 2T$$

$$t =$$

$$t =$$

Trapezoidal Waveform

$$y = mx + b$$

$$i(t) = \begin{cases} \left(\frac{I_p - I_{\min}}{DT} \right) t + I_{\min} & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

I_p

$I_{\min}$

Ripple

Difference
 b/w both
 currents



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Example 2

trapezoid frequency 500 kHz
40% peak current 2 A ripple 300 mA

$$i(t) = \begin{cases} \left(\frac{I_p - I_{min}}{DT} \right) t + I_{min} & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

$$f = 500 \text{ kHz}$$

$$T = \frac{1}{f} = \frac{1}{500 \text{ kHz}} = 2 \mu\text{s}$$

$$D = 40\% = 0.4$$

$$T = 2 \mu\text{s}$$

$$I_p = 2 \text{ A} \quad I_{\text{ripple}} = 300 \text{ mA}$$

$$I_{min} = I_p - I_{\text{ripple}}$$

$$= 2 - 300 \text{ mA}$$

$$I_{min} = 1.7 \text{ A}$$

$$DT = (0.4)(2 \mu\text{s}) = 800 \text{ ns}$$

$$i(t) = \begin{cases} \left(\frac{2 - 1.7}{800 \text{ ns}} \right) t + 1.7 & 0 \leq t < 800 \text{ ns} \\ 0 & 800 \text{ ns} \leq t < 2 \mu\text{s} \end{cases}$$

$$\text{for } t = 400 \text{ ns}$$

$$i(400 \text{ ns}) = 1.85 \text{ A}$$

$$\text{for } t = 1 \mu\text{s}$$

$$i(1 \mu\text{s}) = 0$$

These equations are
valid for 1 cycle

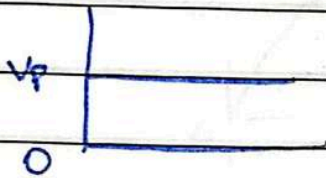


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Average Value

DC waveform

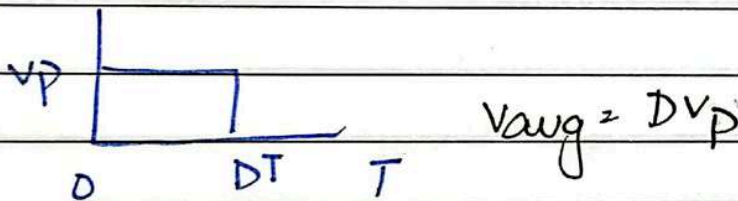


$$V_{ave} = V_p = V_{dc}$$

This waveform is not varying so its peak and average value does not vary

Average value vary if there is change in the waveform

Rectangular Waveform



for $D = 50\%$

$$V_{ave} = \frac{V_p}{2}$$

for $D = 20\%$

$$V_{ave} = \frac{V_p}{5}$$

for $D = 10\%$

$$V_{ave} = \frac{V_p}{10}$$

Here average value is according to the duty cycle

General Formula for Calculating average value

$$V_{avg} = \frac{1}{T} \int_0^T v(t) dt$$

Valid for any signal

→ divided by Time period

for finding average value



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DC waveform

$$\begin{aligned} V(t) &= V_P \\ V_{avg} &= \frac{1}{T} \int_0^T V_P dt \\ &= \frac{1}{T} \int_0^T V_P dt \\ &= \frac{V_P}{T} t \Big|_0^T \\ &= \frac{V_P}{T} [T - 0] \end{aligned}$$

$$\boxed{V_{avg} = V_P}$$

Rectangular Waveform:

$$V(t) = \begin{cases} V_P & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

$$V_{avg} = \frac{1}{T} \left[\int_0^{DT} V_P dt + \int_{DT}^T 0 dt \right]$$

$$= \frac{1}{T} \left[V_P \int_0^{DT} dt + 0 \int_{DT}^T dt \right]$$

$$= \frac{1}{T} \left[V_P (t \Big|_0^{DT}) + 0 \right]$$

$$= \frac{1}{T} [V_P [DT - 0]]$$

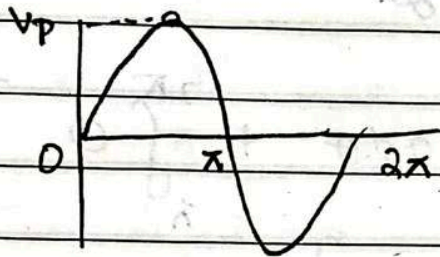
$$\boxed{V_{avg} = V_P D}$$

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Sine wave

Average value of sine wave is 0



$T = 2\pi$ of sine wave

bcz it consist
of both positive &
negative part that
will cancel each
other

$$\begin{aligned} V_{avg} &= \frac{1}{T} \int_0^T v(t) dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} V_p \sin \theta d\theta \\ &= \frac{1}{2\pi} \times V_p \int_0^{2\pi} \sin \theta d\theta \\ &= \frac{V_p}{2\pi} [\cos \theta]_0^{2\pi} \end{aligned}$$

$$= \frac{V_p}{2\pi} [-\cos(2\pi) + \cos(0)]$$

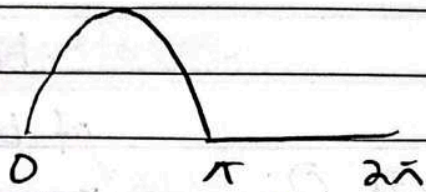
$$= \frac{V_p}{2\pi} [-1 + 1]$$

$$\boxed{V_{avg} = 0}$$



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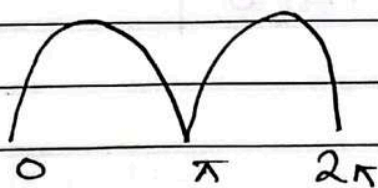
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$$v(t) = \begin{cases} v_p \sin \theta & 0 \leq \theta \leq \pi \\ 0 & \pi \leq \theta \leq 2\pi \end{cases}$$

$$\begin{aligned} v_{avg} &= \frac{1}{T} \int_0^T v(t) dt \\ &= \frac{1}{2\pi} \left[\int_0^\pi v_p \sin \theta d\theta + \int_\pi^{2\pi} 0 d\theta \right] \\ &= \frac{1}{2\pi} \left[v_p \int_0^\pi \sin \theta d\theta \right] \\ &= \frac{1}{2\pi} \left[v_p [-\cos \theta]_0^\pi \right] \\ &= \frac{1}{2\pi} \left[v_p [-\cos(\pi) + \cos(0)] \right] \\ &= \frac{1}{2\pi} v_p [1 + 1] \end{aligned}$$

$$v_{avg} = \frac{v_p}{\pi}$$



$$v(t) = \begin{cases} v_p \sin \theta & 0 \leq \theta \leq \pi \\ -v_p \sin \theta & \pi \leq \theta \leq 2\pi \end{cases}$$

$$\begin{aligned} v_{avg} &= \frac{1}{T} \int_0^T v(t) dt \\ &= \frac{1}{2\pi} \left[\int_0^\pi v_p \sin \theta d\theta - \int_\pi^{2\pi} v_p \sin \theta d\theta \right] \\ &= \frac{1}{2\pi} \left[v_p [-\cos \theta]_0^\pi - v_p [-\cos \theta]_\pi^{2\pi} \right] \end{aligned}$$



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$$V_{avg} = \frac{V_p}{2\pi} [-\cos(\pi) + \cos(0)] - \frac{V_p}{2\pi} [-\cos(2\pi) + \cos(2\pi)]$$

$$= \frac{V_p}{2\pi} [1 + 1] - \frac{V_p}{2\pi} [-1 - 1]$$

$$= \frac{V_p}{\pi} + \frac{V_p}{\pi}$$

$$V_{avg} = \frac{2V_p}{\pi}$$



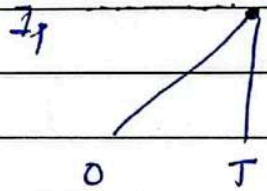
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Lecture 5

Average Value of Sine Wave

Triangular Wave



$$i(t) = \frac{I_p}{T} t$$

$$I_{ave} = \frac{1}{T} \int_0^T \frac{I_p}{T} t dt$$

$$I_{avg} = \frac{1}{T} \int_0^T \frac{I_p}{T} t dt$$

$$= \frac{I_p}{T^2} \int_0^T t dt$$

$$= \frac{I_p}{T^2} \left[\frac{t^2}{2} \right]_0^T$$

$$= \frac{I_p}{2T^2} [T^2 - 0]$$

$$I_{avg} = \frac{I_p}{2}$$

DC signal

$$V_{avg} = V_p$$

Rectangular signal

$$V_{avg} = V_p$$

Sinusoidal

$$V_{avg} = 0$$

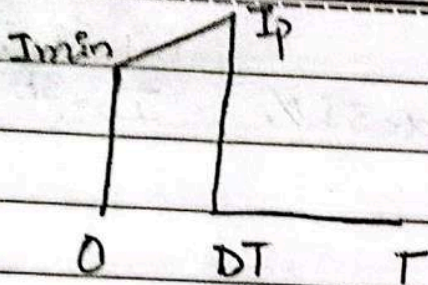
Half wave

$$V_{avg} = \frac{V_p}{\pi}$$

Full wave

$$V_{avg} = \frac{2V_p}{\pi}$$





$$i(t) = \begin{cases} \frac{I_p - I_{min}}{DT} t + I_{min} & 0 \leq t \leq DT \\ 0 & DT < t < T \end{cases}$$

$$i(t) = \frac{(I_p - I_{min})t}{DT} + I_{min}$$

$$I_{avg} = \frac{1}{T} \int_0^{DT} \left(\frac{I_p - I_{min}}{DT} t + I_{min} \right) dt$$

$$= \frac{1}{T} \times \frac{I_p - I_{min}}{DT} \int_0^{DT} t dt + \frac{I_{min}}{T} \int_0^{DT} dt$$

$$= \frac{I_p - I_{min}}{T \times DT} \left[\frac{t^2}{2} \right]_0^{DT} + \frac{1}{T} I_{min} [t]_0^{DT}$$

$$= \frac{I_p - I_{min}}{T \times DT} \left[\frac{D^2 T^2}{2} \right] + \frac{I_{min}}{T} [DT - 0]$$

$$= \left(\frac{I_p - I_{min}}{2} \right) D + I_{min}(D)$$

$$= D \left[\frac{I_p - I_{min} + 2I_{min}}{2} \right]$$

$$= \frac{D}{2} [I_p + I_{min}]$$

$$= \frac{I_p + I_{min} D}{2}$$



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Example 3

frequency = 100 kHz Duty cycle 35% $I_p = 3A$
 $I_{ripple} = 400 mA$

$$I_{avg} = \frac{I_p + I_{min}}{2} D$$

$$I_{min} = I_p - I_{ripple} = 3 - 400 mA = 2.6 A$$

$$= \frac{3 + 2.6}{2} \times 0.35$$

$$I_{avg} = 0.98 A$$

Average value concept
in sine wave is not
meaningful

as it is in both the
+ve & -ve so its avg value
is zero

If not mention ka
Konsa signal then RME
value is taken by default

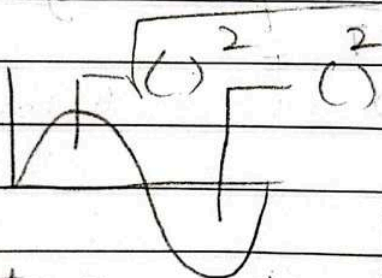
Root Mean Square

↳ meaningful

we have studied ka

$$(1-x)^2 \neq x^2$$

So



to cancel the effect of square so we will take
square root

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$



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DC Signal

$$V_t = V_p$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T V_p^2 dt}$$

$$= \sqrt{\frac{V_p^2}{T} \left(t \right)_0^T}$$

$$= \sqrt{\frac{V_p^2}{T} [T - 0]}$$

$$= \sqrt{\frac{V_p^2}{T} \times T}$$

$$\boxed{V_{rms} = V_p}$$

$$V(t) = V_p = V_{avg} = V_{rms}$$

Rectangular Signal

$$V(t) = \begin{cases} V_p & 0 \leq t < DT \\ 0 & DT \leq t \leq T \end{cases}$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^{DT} V_p^2 dt}$$

$$= \sqrt{\frac{V_p^2 DT}{T} \int_0^{DT} dt}$$

$$= \sqrt{\frac{V_p^2}{T} \left(t \right)_0^{DT}}$$

$$= \sqrt{\frac{V_p^2}{T} \times [DT - 0]}$$



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$$V_{rms} = \sqrt{\frac{V_p^2}{T} \int_0^T dt}$$

$$V_{rms} = V_p \sqrt{\frac{1}{T} \int_0^T dt}$$

$T \neq 0$ cannot exist
at same time

Sine wave

$$V_{rms} = 220V$$

$$V_p = 311V$$

$$v(t) = v_p \sin \theta \quad 0 \leq \theta \leq 2\pi$$

Sinusoidal wave

Same

$$V_{rms} = 220V$$

$$V_{DC} = 220V$$

$$V(t) = \sqrt{\frac{1}{T} \int_0^{2\pi} V_p^2 \sin^2 \theta d\theta}$$

$$= \sqrt{\frac{V_p^2}{T} \int_0^{2\pi} \sin^2 \theta d\theta}$$

$$= \sqrt{\frac{V_p^2}{T} \int_0^{2\pi} (1 - \cos^2 \theta) d\theta}$$

$$= \sqrt{\frac{V_p^2}{T} \left[\frac{\theta}{2} \Big|_0^{2\pi} - \frac{1}{2} \int_0^{2\pi} \cos 2\theta d\theta \right]}$$

$$= \sqrt{\frac{V_p^2}{T} \left[\frac{1}{2} [2\pi - 0] - \frac{1}{2} \left[\frac{\sin 2\theta}{2} \right]_0^{2\pi} \right]}$$

$$= \sqrt{\frac{V_p^2}{T} \left[(\pi) - \frac{1}{2} \left[\frac{\sin 2(2\pi)}{2} - \frac{\sin(0)}{2} \right] \right]}$$

$$= \sqrt{\frac{V_p^2}{T} (\pi)}$$

$$= V_p \sqrt{\frac{\pi}{T}} = \frac{V_p (1.772)}{\sqrt{T}}$$

$$= V_p \sqrt{\frac{\pi}{2\pi}}$$



$$\frac{V}{V_{rms}} = \frac{V_p}{\sqrt{2}}$$

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Half Wave Rectifier

$$v(t) = \begin{cases} v_p \sin \theta & 0 < \theta < \pi \\ 0 & \pi < \theta < 2\pi \end{cases}$$

$$V(t) = \sqrt{\frac{1}{T} \int_0^{\pi} (v_p \sin \theta)^2 d\theta}$$
$$= \sqrt{\frac{v_p^2}{2\pi} \int_0^{\pi} \sin^2 \theta d\theta}$$

$$= \sqrt{\frac{v_p^2}{2\pi} \int_0^{\pi} \frac{1 - \cos 2\theta}{2} d\theta}$$

$$= \sqrt{\frac{v_p^2}{2\pi} \left[\frac{1}{2} \int_0^{\pi} d\theta - \frac{1}{2} \int_0^{\pi} \cos 2\theta d\theta \right]}$$

$$= \sqrt{\frac{v_p^2}{2\pi} \left[\frac{1}{2} [\pi - 0] - \frac{1}{2} \left[\frac{\sin 2\theta}{2} \right]_0^{\pi} \right]}$$

$$= \sqrt{\frac{v_p^2}{2\pi} \left[\frac{1}{2} [\pi] - \frac{1}{2} \left[\frac{\sin(2\pi) - \sin(0)}{2} \right] \right]}$$

$$= \sqrt{\frac{v_p^2}{2\pi} \left[\frac{\pi}{2} \right]} = \sqrt{\frac{v_p^2}{4}}$$

$$V_{avg} = \frac{v_p}{2}$$



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Full Wave Rectifier

$$V(t) = \begin{cases} V_p \sin \theta & 0 < \theta < \pi \\ -V_p \sin \theta & \pi < \theta < 2\pi \end{cases}$$

$$V(t) = \sqrt{\frac{1}{2\pi} \int_0^\pi (V_p \sin \theta)^2 d\theta - \frac{1}{2\pi} \int_\pi^{2\pi} (-V_p \sin \theta)^2 d\theta}$$

$$= \sqrt{\frac{V_p^2}{2\pi} \int_0^\pi 1 - \frac{\cos 2\theta}{2} d\theta - \frac{V_p^2}{2\pi} \int_\pi^{2\pi} 1 - \frac{\cos 2\theta}{2} d\theta}$$

$$= \sqrt{\frac{V_p^2}{2\pi} \left[\frac{1}{2} \int_0^\pi d\theta - \frac{1}{2} \int_0^\pi \cos 2\theta d\theta \right] - \frac{V_p^2}{2\pi} \left[\frac{1}{2} \int_\pi^{2\pi} d\theta - \frac{1}{2} \int_\pi^{2\pi} \cos 2\theta d\theta \right]}$$

$$= \sqrt{\frac{V_p^2}{2\pi} \left[\frac{1}{2} [\pi - 0] - \frac{1}{2} \left[\frac{\sin 2\theta}{2} \right]_0^\pi \right] - \frac{V_p^2}{2\pi} \left[\frac{1}{2} [2\pi - \pi] - \frac{1}{2} \left[\frac{\sin 2\theta}{2} \right]_\pi^{2\pi} \right]}$$

$$= \sqrt{\frac{V_p^2}{2\pi} \left[\frac{\pi}{2} \right] - \frac{V_p^2}{2\pi} \left[\frac{\pi}{2} \right]}$$

$$= \sqrt{\frac{V_p^2}{4} + \frac{V_p^2}{4}}$$

$$= \sqrt{\frac{2V_p^2}{4}}$$

$$= \frac{V_p}{\sqrt{2}}$$



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Example 4

Frequency = 100 kHz

Duty cycle = 35% $V_p = 15 \text{ V}$

Rectangular wave

$$V_{\text{avg}} = V_p \sqrt{D}$$
$$= 15 \sqrt{0.35}$$

$$V_{\text{rms}} = 8.87 \text{ V}$$

Example 5

$f = 60$

$V_p = 170$

Half wave

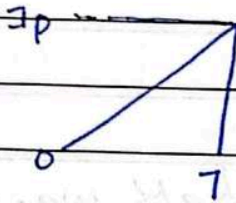
$$V_{\text{rms}} = \frac{V_p}{2} = \frac{170}{2} = 85 \text{ V}$$

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Lecture 6

RMS Value of Triangular Wave



$$i(t) = \frac{I_p}{T} t$$

$$I_{rms}^2 = \frac{1}{T} \int_0^T \left(\frac{I_p}{T} t \right)^2 dt$$

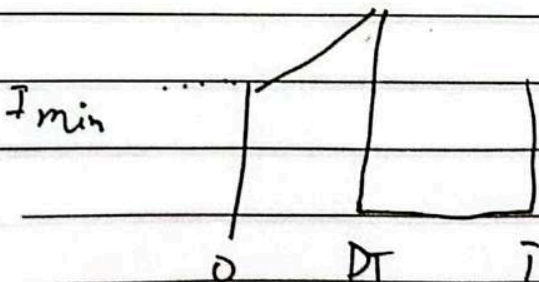
$$= \frac{1}{T} \left(\frac{I_p^2}{T^2} \int_0^T t^2 dt \right)$$

$$= \sqrt{\frac{I_p^2}{T^3} \left(\frac{t^3}{3} \right) \Big|_0^T}$$

$$= \sqrt{\frac{I_p^2 \times T^3}{T^3 \times 3}}$$

$$I_{rms} = \frac{I_p}{\sqrt{3}}$$

RMS Value of Trapezoidal wave



$$i(t) = \begin{cases} \frac{I_p - I_{min}}{DT} t + I_{min} & 0 \leq t < DT \\ I_p & DT \leq t < T \\ 0 & t \geq T \end{cases}$$

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$$I_{rms} = \sqrt{\frac{1}{T} \int_0^{DT} \left(\frac{I_p - I_{min}}{DT} t + I_{min} \right)^2 dt}$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$= \sqrt{\frac{1}{T} \int_0^{DT} \left(\frac{I_p - I_{min}}{DT} t \right)^2 + (I_{min})^2 + 2 \left(\frac{I_p - I_{min}}{DT} t \right) I_{min} dt}$$

$$= \sqrt{\frac{1}{T} \left[\left(\frac{I_p - I_{min}}{DT} \right)^2 \int_0^{DT} t^2 dt + I_{min}^2 \int_0^{DT} dt + 2 \left(\frac{I_p - I_{min}}{DT} I_{min} \right) \int_0^{DT} t dt \right]}$$

$$= \sqrt{\frac{1}{T} \left[\frac{(I_p - I_{min})^2}{D^2 T^2} \frac{t^3}{3} \Big|_0^{DT} + I_{min}^2 t \Big|_0^{DT} + 2 \frac{(I_p - I_{min}) I_{min}}{DT} \frac{t^2}{2} \Big|_0^{DT} \right]}$$

$$= \sqrt{\frac{1}{T} \left[\frac{(I_p - I_{min})^2}{D^2 T^2} \frac{D^3 T^3}{3} + DT I_{min}^2 + 2 I_{min} (I_p - I_{min}) \times \frac{D^2 T^2}{2} \right]}$$

$$= \sqrt{\frac{1}{T} \left[\frac{(I_p - I_{min})^2}{3} \times \frac{DT}{3} + I_{min}^2 DT + I_{min} (I_p - I_{min}) DT \right]}$$

$$= \sqrt{D \left[\frac{(I_p - I_{min})^2}{3} + I_{min}^2 + I_p I_{min} - I_{min}^2 \right]}$$

$$= \sqrt{D \left(\frac{I_p^2 + I_{min}^2 - 2 I_p I_{min} + I_p I_{min}}{3} \right)}$$

$$= \sqrt{D \left(\frac{I_p^2 + I_{min}^2 + I_p I_{min}}{3} \right)}$$

$$I_{rms} = \sqrt{\frac{D}{3} (I_p^2 + I_p I_{min} + I_{min}^2)}$$

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Power

Types of Power

$P_{inst} \rightarrow$ Instantaneous Power

$\rightarrow$ changes w.r.t to time

$$p(t) = v(t) i(t)$$

$\rightarrow$ this gives value at that time

$P_{avg} \rightarrow$ Average Power

$\rightarrow$ exist behind useful work

$$P_{avg} = \frac{1}{T} \int_0^T p(t) dt$$

$\rightarrow$ instantaneous power

DC waveform

$$v(t) = V_P \quad i(t) = I_P$$

$$P = \frac{1}{T} \int_0^T (v(t) i(t)) dt$$

$$= \frac{1}{T} \int_0^T V_P I_P dt$$

$$= \frac{V_P I_P}{T} \left[t \right]_0^T$$

$$P = V_P I_P$$



Rectangular Signal

$$v(t) = \begin{cases} V_p & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

$$i(t) = \begin{cases} I_p & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

$$P = \frac{1}{T} \int_0^{DT} V_p I_p dt$$

$$= \frac{V_p I_p}{T} \int_0^{DT} dt$$

$$= \frac{V_p I_p}{T} [DT]$$

$$|P| = D V_p I_p$$

Sinusoidal Signal

Resistive Load (V and I in phase)

$$v(t) = V_p \sin \theta \quad 0 < \theta \leq 2\pi$$

$$i(t) = I_p \sin \theta \quad 0 < \theta \leq 2\pi$$

$$P = \frac{1}{T} \int v(t) i(t) d\theta$$

$$= \frac{1}{2\pi} \int_0^{2\pi} (V_p \sin \theta)(I_p \sin \theta) d\theta$$

$$= \frac{1}{2\pi} \int_0^{2\pi} V_p I_p \sin^2 \theta d\theta$$



$$= \frac{V_p I_p}{2\pi} \int_0^{2\pi} \sin^2 \theta d\theta$$

$$= \frac{V_p I_p}{2\pi} \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= \frac{V_p I_p}{2\pi} \left[\frac{1}{2} \int_0^{2\pi} d\theta - \frac{1}{2} \int_0^{2\pi} \cos 2\theta d\theta \right]$$

$$= \frac{V_p I_p}{2\pi} \left[\frac{1}{2} [2\pi - 0] - \frac{1}{2} \left[\frac{\sin 2\theta}{2} \right]_0^{2\pi} \right]$$

$$= \frac{V_p I_p}{2\pi} [\pi - 0]$$

P = $\frac{V_p I_p}{2}$

we are using peak value $V_{rms} = \frac{V_p}{\sqrt{2}}$

$$V_p = \sqrt{2} V_{rms}$$

$$P = \frac{\sqrt{2} V_{rms} \sqrt{2} I_{rms}}{2} = V_{rms} I_{rms}$$

Half wave

$$V(t) = \begin{cases} V_p \sin \theta & 0 < \theta < \pi \\ 0 & \pi < \theta < 2\pi \end{cases}$$

$$I(t) = \begin{cases} I_p \sin \theta & 0 \leq \theta < \pi \\ 0 & \pi \leq \theta < 2\pi \end{cases}$$

$$P = \frac{1}{T} \int V(t) I(t) d\theta$$

$$= \frac{1}{2\pi} \int_0^{\pi} (V_p \sin \theta) (I_p \sin \theta) d\theta$$

$$= \frac{I_p V_p}{2\pi} \int_0^{\pi} \sin^2 \theta d\theta$$

$$= \frac{I_p V_p}{2\pi} \int_0^{\pi} \frac{1 - \cos 2\theta}{2} d\theta$$

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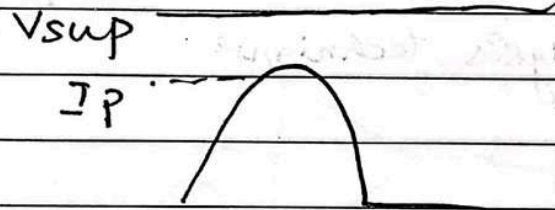
$$= \frac{I_p V_p}{2\pi} \left[\frac{1}{2} \int_0^{\pi} d\theta - \frac{1}{2} \int_0^{\pi} \cos 2\theta d\theta \right]$$

$$= \frac{I_p V_p}{2\pi} \left[\frac{1}{2} [\pi - 0] - \frac{1}{2} \left[\frac{\sin 2\theta}{2} \right]_0^{\pi} \right]$$

$$P = \frac{I_p V_p}{4}$$

Example 1

Average power



$$v(\theta) = V_{sup}$$

$$i(\theta) = \begin{cases} I_p \sin \theta & 0 < \theta < \pi \\ 0 & \pi < \theta < 2\pi \end{cases}$$

$$P = \frac{1}{2\pi} \left[\int_0^{\pi} V_{sup} I_p \sin \theta d\theta \right]$$

$$= \frac{1}{2\pi} \left[V_{sup} I_p \int_0^{\pi} \sin \theta d\theta \right]$$

$$= \frac{1}{2\pi} \left[V_{sup} I_p \left[-\cos \theta \right]_0^{\pi} \right]$$

$$= \frac{V_{sup} I_p}{2\pi} [-\cos \pi + \cos 0]$$

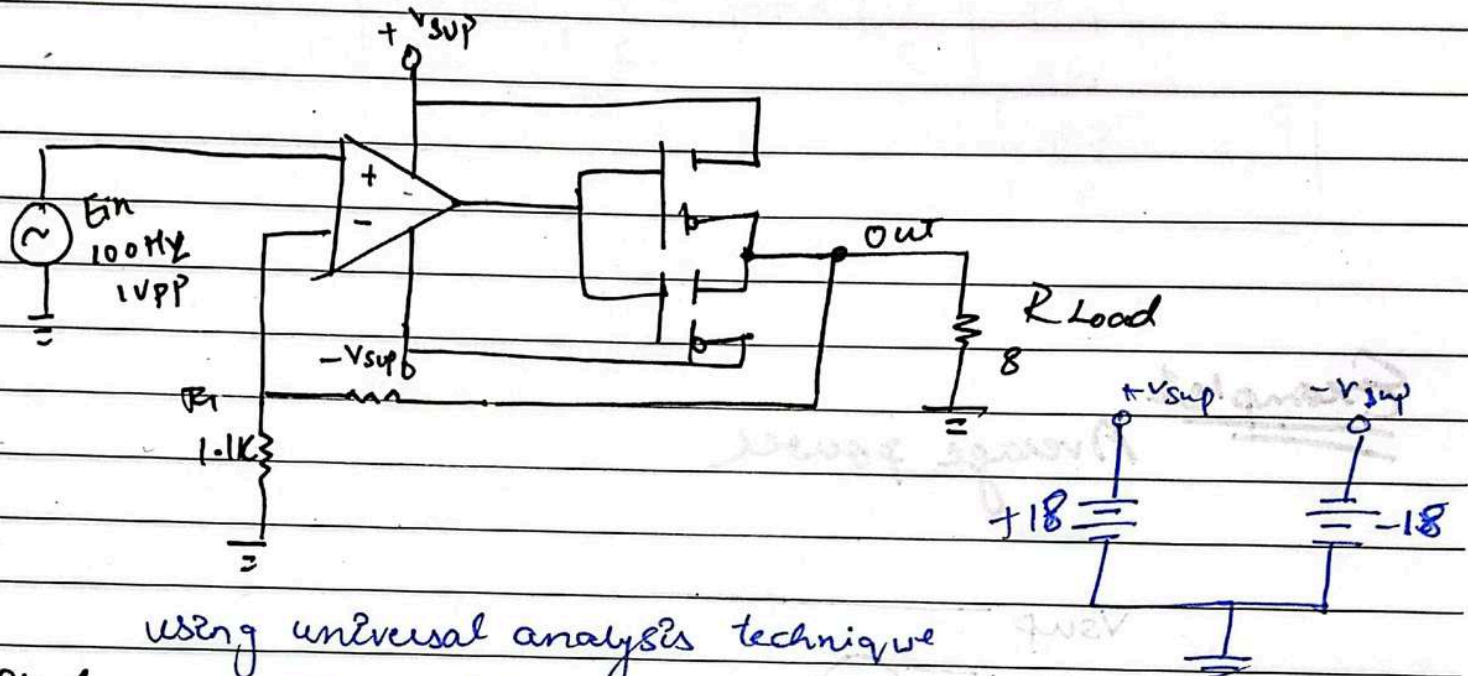
$$P = \frac{V_{sup} I_p}{\pi}$$

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Example 2



using universal analysis technique

Step 1

$$V_{NI} = 1V_p$$

Step 2

$$V_{IN} = 1V_p$$

Step 3

$$I_{in} = \frac{1}{1.1K} = 0.909mA$$

Step 4

$$V_{out} = (0.909m)(10 + 1.1K) = 10.08$$

Step 5

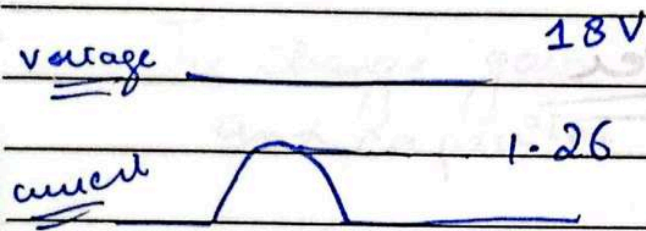
$$I_{Load} = \frac{10.08}{8} = 1.26V$$

$$P_{out} = (V_{out})(I_{Load}) = (10.08 \times 1.26) = 6.35W$$

(2) $\rightarrow$ bcr peak value



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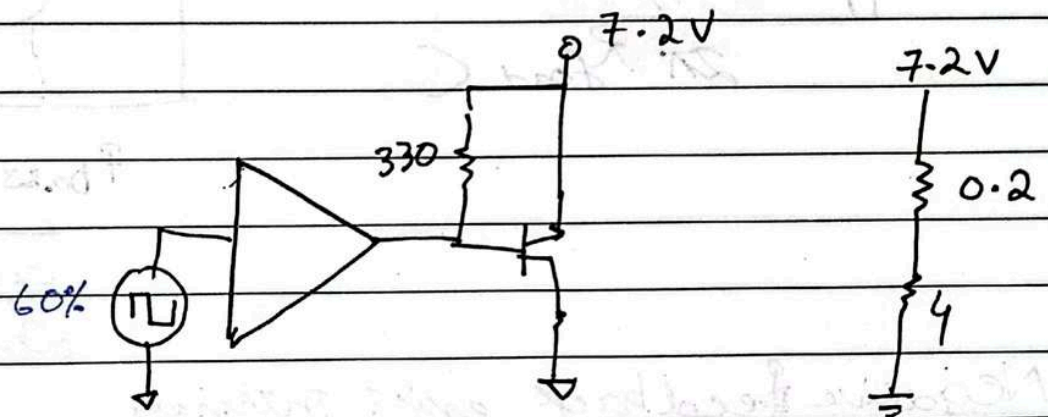


$$P_{\text{sup}} = V_{\text{sup}} I_p = 18(1.26) = 7.22$$

$$\text{Total } p_{\text{sup}} = 7.22 + 7.22 = 14.44$$

→ used is 6.35W remain is use for heating IC

Example 4



$$I_z = \frac{7.2 - 0}{0.2 + 4} = 1.71$$

$$V_{\text{Load}} = I_z R = (1.71)(4) = 6.85$$

$$P = VI = (1.71)(6.85)(0.6)$$

$$P_{\text{Load}} = 7.028W$$

$$V_{\text{tran}} = 0.2(1.71) = 0.342$$

$$P_{\text{Trans}} = 0.6(0.342)(1.71)$$

$$P_{\text{Trans}} = 0.3508W$$



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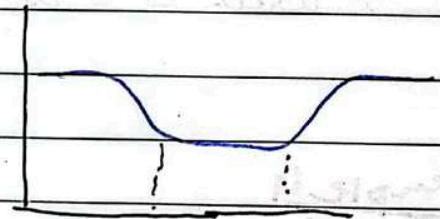
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Low Power Audio Amplifier

- operated by single supply
↳ +ve supply but not always +ve voltage
- this amplifier could have negative output

$$V_{pp} \approx V_{CC}$$

$$f_1 = \frac{1}{2\pi R_{load} C}$$



f_{bass} f_{high}

cut off frequency
Lower & higher frequency

- Negative feedback exist internally
 $R_f = 15K$ b/w 1 & 5

- No large output impedance (50k)

↳ that mean current can pass

↳ To solve this problem we add a potentiometer to stop current flowing inside op amp

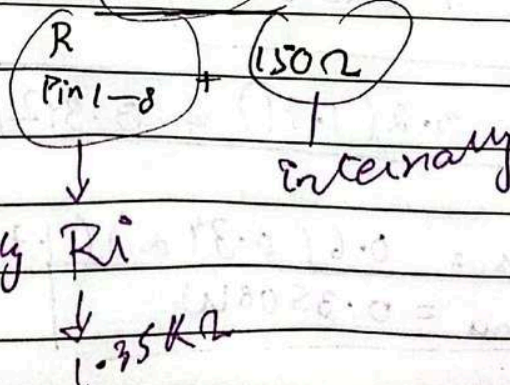
$$A \approx 2 \times (15K\Omega) \rightarrow R_f$$

Ideal gain
of IC is 20

↳ Inverting → ground



↳ Non inverting → supply



To change gain we need to apply resistor and capacitor b/w 1-8

If only C is applied

$$A = 2 \times \frac{150k\Omega}{150}$$

$$A = 200 \rightarrow \text{max gain}$$

$$\left| C > \frac{1}{2\pi f \left\{ \frac{R}{f} \right\}} \right| \rightarrow \text{capacitive effect}$$

Example

gain 40 $f_i = 80\text{Hz}$

$$40 = 2 \times 15k\Omega$$

$$R_{pin} + 150$$

$$40 (R_{pin} + 150) = 30k\Omega$$

$$R_{pin} + 150 = \frac{30k\Omega}{40}$$

$$R_{pin} = 750 - 150$$

$$= 600 \text{ not available take } 620$$

$$C = \frac{1}{2\pi f \left(\frac{R}{f} \right)} = 22.46\text{ nF commercial available } 27\text{ nF}$$

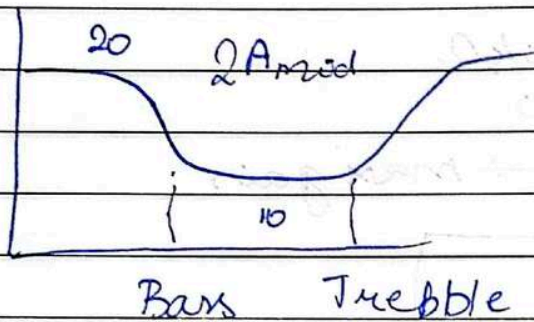


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Lecture 7

Low Power audio amplifier

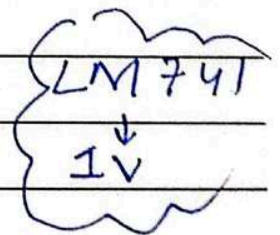


- ↳ Low frequency behaves as DC
- ↳ and capacitor block DC and act as open circuit
- ↳ max gain 20

for bass pin 1 and 5
for treble pin 1 and 8

C treble = $\frac{1}{2\pi R_{treble} \times f_{treble boost}}$

Low power



OPA548 operational Amplifier

- ↳ Power amplifier → can supply external power
- ↳ When power increases they exist a mismatch b/w V_t & V_- called V_{offset}



- ↳ Input current is not zero
- ↳ Input impedance is only 50k

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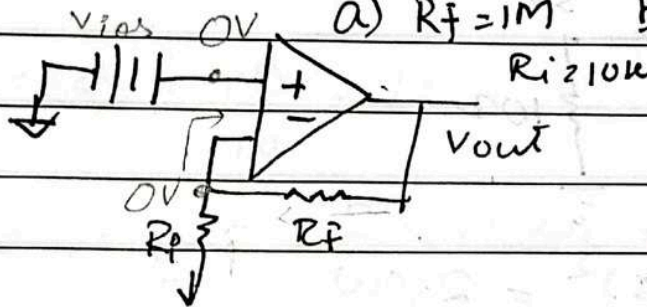
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→ Input offset current 500 nA

Example 1

$$V_{ios} = 0\text{ V} \quad I_{in2\text{ pin } 2} = 500\text{ nA}$$

$$\text{a) } R_f = 1\text{ M} \quad \text{b) } R_f = 10\text{ k}\Omega \quad R_i = 100\Omega$$



$$I_{in2\text{ pin } 2} = 500\text{ nA}$$

$$I_i = 0 \quad V_i = 0\text{ V}$$

By applying KCL

$$I_{in2\text{ pin } 2} = I_i + I_f$$

$$I_i = 0$$

$$I_{in2\text{ pin } 2} = I_f = 500\text{ nA}$$

$$V_{out} = I_f R_f = (500\text{ nA})(1\text{ M}) = 0.5\text{ V}$$

$$\text{if } R_f = 10\text{ k}$$

$$V_{out} = (500\text{ nA})(10\text{ k}) = 5\text{ mV}$$

Power Calculations with DC signal

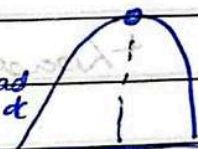
$$P_{Load} = V_{load} I_{load}$$

$$\therefore I_{supply} = I_{load}$$

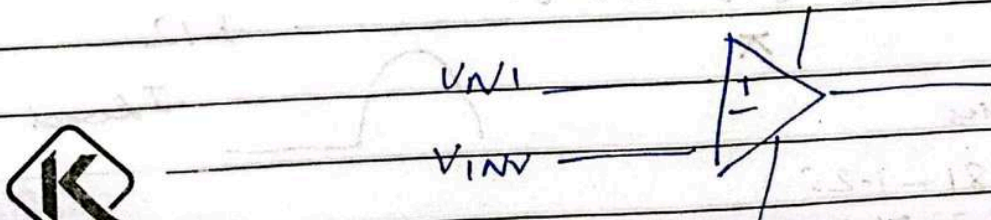
$$P_{supply} = V_{supply} I_{supply}$$

$$P_{supply} = V_{supply} I_{load\text{dc}}$$

$$P_{ic} = P_{supply} - P_{Load}$$



max
power
dissipated
at $\frac{1}{2} V_{supply}$



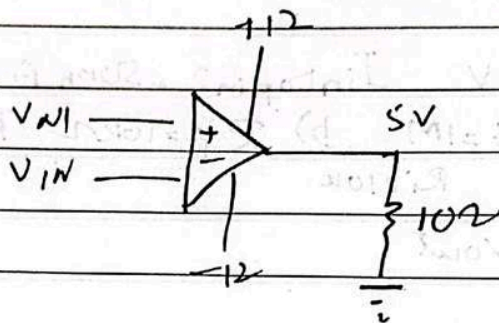
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Example 3

$\pm 12\text{ Vdc}$ delivering $+5\text{ Vdc}$ 10Ω resistive load



$$P_{\text{load}} = \frac{V^2}{R} = \frac{(5)^2}{10} = 2.5\text{ W}$$

$$P_{\text{supply}} = 12 \left(\frac{5}{10} \right) = 6\text{ W}$$

$$P_{\text{IC}} = 6 - 2.5 = 3.5\text{ W}$$

Q. If supply become 9

→ Load same
 → IC dissipation reduce

Power Calculation with Sinusoidal

Example 4

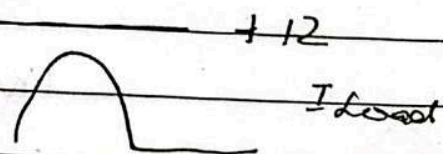
$$P_{\text{load}} = \frac{V^2}{R} = \frac{(5/\sqrt{2})^2}{10} = 1.25\text{ W}$$

$$I_{\text{load}} = \frac{V}{R} = 0.5\text{ A}$$

$$P_{\text{load}} = \frac{V_p I_p}{2} = \frac{5 \times 0.5}{2} = 1.25\text{ W}$$

$$P_{\text{supply}} = 2 \times 12\text{ Vdc} \times \frac{0.5}{\pi} = 3.81$$

↓
 by 2 supplies



$$P_{\text{IC}} = 3.81 - 1.25 = 2.5\text{ W}$$

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When voltage 10

Voltage double
power 4

$$P_{\text{Load}} = \frac{V^2}{R} = \frac{(10)^2}{10} = 10 \text{ W}$$

Time increases

$$I_{\text{Load}} = \frac{V}{R} = 1 \text{ A}$$

$$P_{\text{Supply}} = 2 \times 12 \text{ Vdc} \times \frac{10 \text{ W}}{1 \text{ V}} = 240 \text{ W}$$

$$P_{\text{IC}} = 7.6 - 5$$

$$P_{\text{IC}} = 2.6$$

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Lecture 8

OPA548 Operational Amplifier

Gain bandwidth product is typically 1
 $GBW = A_C \times f_H$

A_o

↳ open loop gain

$$A_o = \frac{V_o}{V_i}$$

Slew rate

Max slew rate OPA548 is 10V/μs



Sharp rise & fall
max change

Example 2

$V_i = \text{Input} = 300 \text{ mV}_{\text{rms}}$

SA of current ability to pass

$P_{\text{load}} = 35 \text{ W}$

$R_{\text{load}} = 8 \Omega$

$$P = \frac{V^2}{R}$$

$$P = i^2 R$$

$$35 = i^2 (8)$$

$$35 = \frac{V^2}{8}$$

$$i = 2.09 \text{ Arms} = 3 \text{ Ap}$$

$$V = 16.73 \text{ V}_{\text{rms}} = 23.66 \text{ V}_p$$

$$f = \frac{SR}{2\pi V_p}$$

$$= \frac{10 \text{ V}/\mu\text{s}}{2\pi (23.66)}$$

$$= 67.267 \text{ kHz}$$

can be use for audio 20Hz - 20kHz range



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$$A_{O2} = \frac{V_O}{V_i} = \frac{16.76}{300m}, \frac{23.66}{\sqrt{2} 300m}, 55.76$$

$$GBW = 55.6 \times 20k$$

$$= 1.1 MHz$$

max range

of audio amplifier

low range amplifier

56 kHz

Not good amplifier medium range amplifier

medium range amplifier

1-5 MHz

High Range > 5 MHz



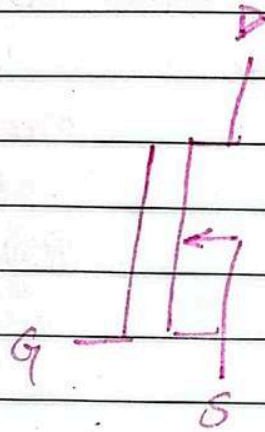
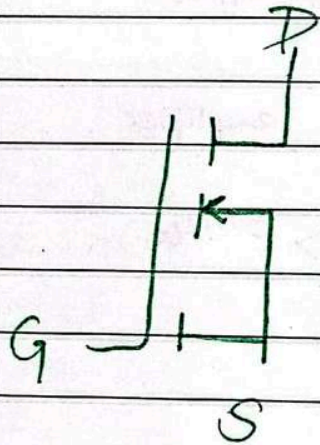
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Lecture 8

Enhancement Mode MOSFET



- Complete channel
- By default on
- Can be off by applying -ve voltage on gate

Enhancement
Type

Depletion
Mode

- Normally off (no current flow when supply is given)
- It will be on by applying positive trigger or pulse at gate
- Use in power electronic

Drain → n type → electrons → -ve charge
Source ↓

Gate ↓ → p type → holes

Silicon dioxide glass → insulator → stop following of current

Gate voltage V_{GS}

Threshold voltage V_{th}

$$V_{GS} > V_{th}$$

the voltage increase
more conduction

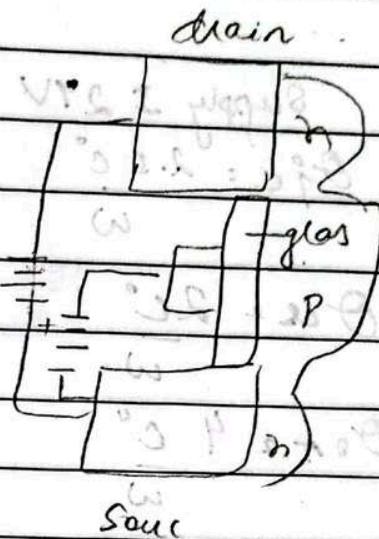
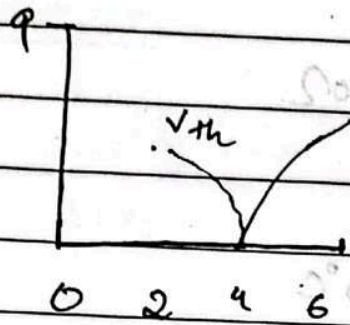


↓
Condition
for conduction

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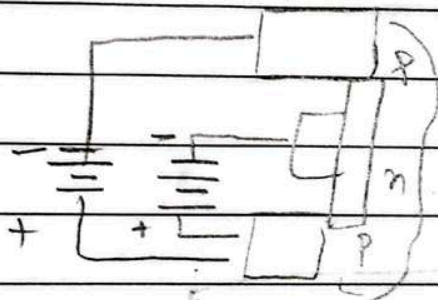
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n channel

P type channel



threshold -ve

current flowing is called drain current

Heat sink is necessary

$$T_d = T_A + P (\theta_{jc} + \theta_{cs} + \theta_{sa})$$

$\downarrow$ allowed Temp $\downarrow$ surrounding Temp $\downarrow$ Thermal Resistance $\downarrow$ case sink $\downarrow$ sink & ambient

$\downarrow$ if resistance of material is high
 so it will not heat up

Normally $T \leq 25^\circ\text{C}$

More cold more life



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Example

Supply $\pm 27V$ $R_{load} = 10\Omega$
 $\theta_{jc} = \frac{2.5^\circ}{W}$

$\theta_{cs} = \frac{2^\circ}{W}$

$\theta_{sa} = \frac{4^\circ}{W}$

$T_A = 40^\circ C$

$T_j \leq 125^\circ C$

$$T_j = T_A + P(\theta_{jc} + \theta_{cs} + \theta_{sa})$$
$$125 = 40 + P(2.5 + 2 + 4)$$

$$85 = P(8.5)$$

$$P = 10W$$

$$P = i^2 R$$

$$10 = i^2$$

$$10$$

$$i = 1A$$

$$P_{jc} = P_{supply} - P_{load}$$

don't use here



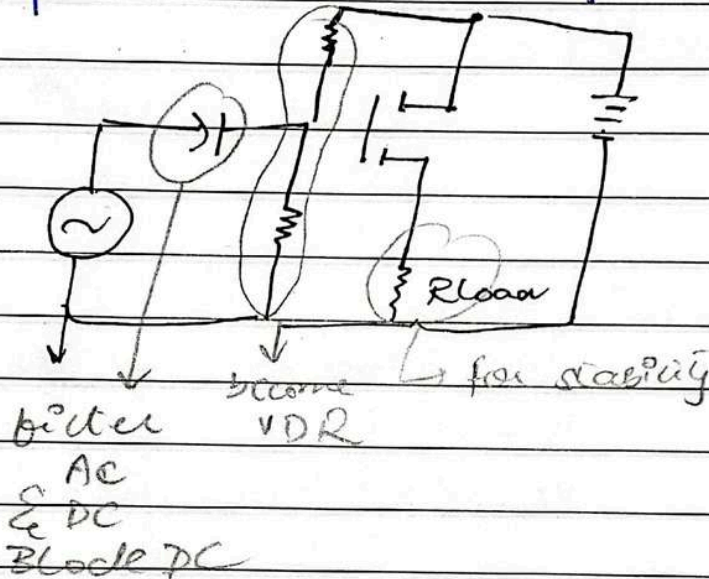
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Lecture 9

Class Amplifier / Common Drain Amplifier

Input reproduced at the output



$$g_{fs} = \frac{\Delta I_D}{\Delta V_{GS}} \quad \left| \begin{array}{l} V_{DS} \text{ constant} \end{array} \right.$$

IRF530 → common drain μ_n is greater than 6.4 A/V
1V change give 6.4V

$$V_{load} = \frac{g_{fs} R_{load}}{1 + g_{fs} R_{load}} \times V_{in}$$

$$I_D = g_{fs} V_{GS}$$

for finding V_{in} we can apply Loop

$$-V_{in} + V_{GS} + V_{load} = 0$$

$$V_{in} = V_{GS} + V_{load}$$



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$$e_{in} = V_{GS} + i_D R_{Load}$$

$$= V_{GS} + g_{fs} V_{GS} R_{Load}$$

$$e_{in} = V_{GS} (1 + g_{fs} R_{Load})$$

$$V_{GS} = \frac{e_{in}}{1 + g_{fs} R_{Load}}$$

$$V_{Load} = i_D R_{Load}$$

$$= g_{fs} V_{GS} R_{Load}$$

$$V_{Load} = \left\{ \frac{g_{fs} R_{Load}}{1 + g_{fs} R_{Load}} \right\} \times e_{in}$$

→ always less than 1

- These amplifier gain is always less 1 whatever input is on output it will not amplify
- These provide large current to R_{Load}
- Worst amplifier

Example $R_2 = 30k\Omega$ $V_{DC} = 56V$ $e_{in} = 20V$

$$V_{GS} = 4.4V_{DC}$$

$$e_{in} = 20V_{p} = 14.14V_{rms}$$

$$V_{G_{DC}} = \left(\frac{30}{30 + 22} \right) 56 = 32.33V$$

$$V_{Load} = 32.33 - 4.4$$

$$DC = 27.9 \approx 28V$$

$$A < 1 \approx 0.98$$



$$\frac{V_{Load}}{e_{in}} = \text{Input (gain)}$$

$$= 0.98 (31.08V) = 31.68V$$